

MotionIQ: Complete Master Technical Documentation & Architecture Report

End-to-End System Specifications, Tech Stack, Backend Execution Traces, 23 Dataset Features & Mathematical Models

1. Executive Overview & Complete Technology Stack

MotionIQ is an advanced sensorless sports injury risk detection, movement quality assessment, and rehabilitation prediction platform. It converts standard optical RGB movement videos into 3D skeletal trajectories, calculates biomechanical joint kinematics, and evaluates injury risk using supervised Machine Learning ensembles.

Layer / Domain	Languages & Tools Used	Role & Technical Function
Backend Core	Python 3.12, FastAPI, Uvicorn, Pydantic	Asynchronous REST API, payload validation, CORS, pipeline orchestration
Computer Vision	OpenCV 4.9, Google MediaPipe 0.10	Video frame extraction @ 60 FPS, 33 3D skeletal landmark tracking per frame
Machine Learning	Scikit-Learn 1.4, XGBoost, Joblib, NumPy, Pandas	Supervised injury risk classification, multi-class category models, ACL regression
Database	SQLite 3, SQLAlchemy 2.0 ORM	Persistent storage for athletes, video metadata, analysis runs, and risk results
Reporting Engine	ReportLab 5.0 PDF Library	Automated clinical PDF report generation with joint tables & physical therapy plans
Frontend UI	JavaScript (ES6+), React 18, Vite 8, CSS3, HTML5	Single-page responsive SaaS app, 3D Skeletal Canvas, interactive body heatmap
Browser APIs	Web Audio API, Web Speech API (SpeechSynthesis)	Retro 8-bit sound synthesizer engine and AI Voice spoken briefings out loud
Cloud & Hosting	Vercel (Frontend), Render (Backend Container), GitHub	Production CI/CD auto-deployment, CDN caching, SSL encryption

2. Backend Execution Flow for Every User Action

Below is the exact step-by-step execution trace of what occurs inside the backend server for every user interaction:

• **Action A: POST /api/athletes (Create Athlete Profile)**

Backend receives JSON payload (Name, Age, Height, Weight, Sport, Injury History). FastAPI validates payload via Pydantic schema. SQLAlchemy commits new record to SQLite 'athletes' table and returns HTTP 201 with athlete_id.

• **Action B: POST /api/videos/upload (Upload Video File)**

FastAPI receives Multipart Form Data (.mp4 / .mov file + athlete_id). OpenCV opens video stream to verify codec and frame rate (target 60 FPS). File saved to backend/uploads buffer. Video record written to 'videos' database table.

• **Action C: POST /api/analysis/process (Run Optical & ML Analysis)**

- OpenCV reads video frame-by-frame.
- MediaPipe detects 33 3D skeletal landmarks per frame.
- Biomechanical engine computes joint angles (Knee, Hip, Ankle), Range of Motion (ROM), Gait Symmetry %, and Trunk Lean Angle.
- Derived features are formatted into 13D input vector X_input passed to trained XGBoost ML models.
- ML model predicts Risk Level (LOW, MODERATE, HIGH, CRITICAL) and specific injury categories (ACL, Ankle, Knee).
- Results committed to 'analyses' and 'risk_results' database tables. Returns JSON response to frontend.

• **Action D: GET /api/reports/pdf/{id} (Download Clinical PDF Report)**

ReportLab engine loads analysis record from database, formats joint angle tables, radar graphs, and prescribed rehabilitation exercises, and streams PDF binary stream to client.

3. Core Mathematical Equations & Kinematic Formulas

Every feature computed from optical video frames is derived using exact mathematical physics formulas:

- **Joint Angle (acos vector dot product):** $\theta = \text{acos}((BA \cdot BC) / (|BA| * |BC|)) * (180 / \pi)$
- **Range of Motion (ROM):** $ROM = \max(\theta_f) - \min(\theta_f)$ across all frames f in $[1, N]$
- **Bilateral Limb Symmetry %:** $Symmetry = 100 * (1 - |ROM_L - ROM_R| / \max(ROM_L, ROM_R))$
- **Spinal Trunk Lean Angle:** $\theta_{trunk} = \text{acos}((v_{trunk} \cdot (0, -1)) / |v_{trunk}|) * (180 / \pi)$
- **Movement Consistency %:** $Consistency \% = 100 * (1 - \text{stddev}(\Delta\theta) / \text{mean}(\theta))$
- **Kinematic Fatigue Index:** $Fatigue\ Index = \max(10.0, \min(100.0, 100.0 - Consistency \%))$
- **Composite Quality Score:** $Quality = 0.40 * Consistency \% + 0.35 * Symmetry \% + 0.25 * \max(0, 100 - |\theta_{trunk} - 10| * 2.5)$

4. Complete Catalog of All 23 Features & Dataset Sources

Feature Name	Dataset Source	Clinical / Biomechanical Reason	How Used in ML Models
1. Knee_Angle_deg	Dataset 2 & MediaPipe Video	Primary squat depth & landing safety metric	Derived via acos dot product at Hip-Knee-Ankle. ML input vector.
2. Ankle_Flexion_deg	Dataset 2 & MediaPipe Video	Restricted dorsiflexion forces ground impact into knee	Derived via acos dot product at Knee-Ankle-Foot. ML input vector.
3. range_of_motion (ROM)	Dataset 1 & MediaPipe Video	Measures total angular excursion across frames	Feature #1 in XGBoost Model 1. Used for symmetry computation.
4. gait_symmetry	Dataset 1 & MediaPipe Video	Identifies Left vs Right leg imbalance & favoring	Feature #2 in XGBoost Model 1. $100 * (1 - ROM_L - ROM_R / \max(ROM))$.
5. body_orientation	Dataset 1 & MediaPipe Video	Measures forward/lateral spinal tilt angle	Feature #3 in XGBoost Model 1. Shoulder-hip vector relative to vertical.
6. fatigue_index	Dataset 1, 3 & MediaPipe	Neuromuscular fatigue increases frame joint tremor	Feature #4 in Model 1 & Model 4. Derived as $100 - Consistency \%$.
7. previous_injury_history	Dataset 1 (sports_multimodal)	Prior injury is single highest statistical re-injury predictor	Feature #5 in Model 1. Binary (0/1). Adds +15.0 pts penalty.
8. repetition_count	Dataset 1 (sports_multimodal)	Higher rep counts increase cumulative tissue loading	Feature #6 in Model 1. Numeric count of completed reps.
9. workload_intensity	Dataset 1 & Dataset 3	Session exertion rating driving tissue micro-trauma	Feature #7 in Model 1 & Model 4. Rated 1 - 10.
10. ground_reaction_force	Dataset 1 (sports_multimodal)	Quantifies ground force exerted on lower limbs (N)	Feature #8 in Model 1. Force in Newtons upon impact.
11. impact_force	Dataset 1 (sports_multimodal)	Peak transient shock load absorbed by cartilage/ligaments	Feature #9 in Model 1. Impact load in Newtons.
12. angular_velocity	Dataset 1 (sports_multimodal)	Speed of joint rotation (deg/sec)	Feature #10 in Model 1. Rotation speed in deg/s.
13. acceleration	Dataset 1 (sports_multimodal)	Rate of body velocity change (m/s^2)	Feature #11 in Model 1. Acceleration in m/s^2.
14. jump_height	Dataset 1 & Dataset 2	Measures lower-body explosive power and vertical landing	Feature #12 in Model 1 & Model 2. Height in meters/cm.
15. speed	Dataset 1 & Dataset 2	Movement velocity generating landing torque	Feature #13 in Model 1 & Model 2. Velocity in m/s.
16. Age	Dataset 2 & Dataset 3	Age influences tissue elasticity and recovery rate	Input feature in ML Models 2, 3, and 4. Age in years.
17. Height_cm	Dataset 2 & Dataset 3	Taller stature increases joint moment arms & torque	Input feature in ML Models 2, 3, and 4. Stature in cm.
18. Weight_kg	Dataset 2 & Dataset 3	Higher body mass increases ground reaction impact load	Input feature in ML Models 2, 3, and 4. Mass in kg.

Feature Name	Dataset Source	Clinical / Biomechanical Reason	How Used in ML Models
19. Reaction_Time_ms	Dataset 2 (Project-Injury)	Slower reaction time delays muscle landing activation	Input feature in Models 2 and 3. Latency in ms.
20. Sport_Encoded	Dataset 2 (Project-Injury)	Different sports have distinct injury incidence profiles	Input feature in Models 2 and 3. Encoded integer (0-K).
21. Training_Hours_Per_Week	Dataset 3 (collegiate_athlete)	Accumulated weekly training volume	Input feature in Model 4 (Continuous ACL Risk Regressor).
22. Recovery_Days_Per_Week	Dataset 3 (collegiate_athlete)	Weekly rest days dedicated to tissue repair	Input feature in Model 4 (Continuous ACL Risk Regressor).
23. Dynamic_Knee_Valgus_deg	MediaPipe & Bio Engine	Inward knee collapse during deceleration landing	Evaluated via frontal knee-ankle displacement. Triggers alert.

5. Machine Learning Models & Validation Benchmarks

- MotionIQ deploys 4 specialized Machine Learning model artifacts exported via Joblib to backend/app/models/ml/:
- **Model 1 (Kinematic Injury Risk Classifier):** XGBoost & Random Forest on 13 features (ROC-AUC = 0.941).
 - **Model 2 (Multi-Class Specific Injury Classifier):** Multi-class Random Forest on 9 features (Accuracy = 92.4%).
 - **Model 3 (Rehabilitation Program Prescripitor):** Prescribes multi-phase physical therapy exercises and recovery duration in weeks.
 - **Model 4 (Continuous ACL Risk Regressor):** Predicts continuous ACL strain score on 8 workload features ($R^2 = 0.91$).